BEV Auxiliary Motor Counts by Vehicle Segment (2024-2026)

This document provides an estimated quantification of auxiliary electric motors across European and Asian BEV models sold in Europe. The estimates are derived from documented features of representative vehicles and typical motor applications identified in prior research.

Motor Categories:
* **Small Stepper Motors (<50W):** Precision actuators (e.g., HVAC flaps, headlamp leveling).
* **Medium DC Motors (50-500W):** Higher power continuous-motion (e.g., pumps, window lifts, seat adjustment).
* **Medium Stepper Motors (50-200W):** Niche applications, assumed to be minimal (0) in this analysis.

A-Segment: Mini Cars

- **Representative Models:** Fiat 500e, Mini Cooper Electric
- **Segment Characteristics:** Basic feature set, focused on efficiency and cost. Minimalist comfort and convenience features.

Motor Type	Estimated Count	Basis for Estimate (Documented Features) -
Small Stepper Motors	**~6-8**	HVAC (4-6 units for blend/mode/recirculation), Headlamp Leveling (2 units), Charging Port Lock (1 unit, if motorized). -
Medium DC Motors	**~3-5**	Window Lifts (2 units), Windshield Wipers (1-2 units), Coolant Pump (1 unit). -
Medium Stepper Motors	**0**	No common applications identified. -
Total per Vehicle	**~9-13**	-

B-Segment: Small Cars

- **Representative Models:** VW ID.3, Peugeot e-208, Opel Corsa-e
- **Segment Characteristics:** More comfort features than A-segment, such as rear power windows and more sophisticated climate control.

Motor Type	Estimated Count	Basis for Estimate (Documented Features) -
Small Stepper Motors	**~8-10**	HVAC (4-6 units), Headlamp Leveling (2 units), Active Grille Shutter (1-2 units), Charging Port Lock (1 unit). -
Medium DC Motors	**~6-8**	Window Lifts (4 units), Windshield Wipers (1-2 units), Coolant Pump (1 unit), Power Folding Mirrors (2 units). -
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Medium Stepper Motors	**0**	No common applications identified. -
Total per Vehicle	**~14-18**	-

C-Segment: Compact Cars

- **Representative Models:** Tesla Model 3, Nissan Ariya, Cupra Born
- **Segment Characteristics:** Advanced features become more common, such as panoramic roofs and basic power seat adjustments.

Motor Type	Estimated Count	Basis for Estimate (Documented Features) -
Small Stepper Motors	**~10-14**	HVAC (6-8 units for dual-zone), Headlamp Leveling/ Matrix LED (2-4 units), Active Grille Shutter (1-2 units), Charging Port Lock (1 unit). -
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Medium DC Motors	**~10-16**	Window Lifts (4 units), Windshield Wipers (1-2 units), Coolant Pumps (1-2 units), Power Folding Mirrors (2 units), Basic Power Seats (2-4 motors for 4/6-way adjustment), Sunroof (1-2 units). -
Medium Stepper Motors	**0**	No common applications identified. -
Total per Vehicle	**~20-30**	-

D-Segment: Mid-Size Cars

- **Representative Models:** Tesla Model S, BMW i4, Polestar 2
- **Segment Characteristics:** High-end features are standard, including advanced driver-assistance systems (ADAS), premium audio, and extensive power seat adjustments.

Motor Type	Estimated Count	Basis for Estimate (Documented Features) -
Small Stepper Motors	**~12-18**	HVAC (6-8 units), Adaptive Headlamps (4 units), Active Grille Shutter (2 units), Charging Port Lock (1-2 units), Power-folding mirrors (2 units). -
Medium DC Motors	**~15-25**	Window Lifts (4), Wipers (2), Coolant Pumps (2), Power Seats (8-12 for 12/16-way adjustment), Sunroof (2), Power Liftgate (2), Soft-close doors (4, optional). -
Medium Stepper Motors	**0**	No common applications identified. -
Total per Vehicle	**~27-43**	-

E-Segment: Executive Cars

- **Representative Models:** Mercedes EQS, BMW iX, Audi e-tron GT
- **Segment Characteristics:** Luxury-focused with extensive comfort and convenience features, such as massaging seats, advanced climate control, and automatic doors.

Motor Type	Estimated Count	Basis for Estimate (Documented Features) -
Small Stepper Motors	**~15-25**	HVAC (8-12 units for multi-zone), Adaptive Headlamps (4-6 units), Active Aero components (2-4 units), Charging Port Locks (2 units), Automatic Doors (4 units, optional). -
Medium DC Motors	**~25-40**	Window Lifts (4), Wipers (2), Coolant Pumps (2-3), Massaging/Ventilated Seats (10-16), Sunroof (2), Power Liftgate (2), Soft-close doors (4), Power steering column (2), Power-deployable door handles (4). -
Medium Stepper Motors	**0**	No common applications identified. -
Total per Vehicle	**~40-65**	-

F-Segment: Luxury Cars

- **Representative Models:** Porsche Taycan, Mercedes EQS SUV, BMW iX xDrive50
- **Segment Characteristics:** Pinnacle of luxury and technology. Features often include everything from the E-segment plus more bespoke and elaborate motorized systems.

Motor Type	Estimated Count	Basis for Estimate (Documented Features) -
Small Stepper Motors	**~20-30+**	HVAC (10-15 units for 4/5-zone), Advanced Adaptive Headlamps (6-8 units), Multiple Active Aero components (4-6 units), Charging Port Locks (2 units), Automatic Doors (4 units). -
Medium DC Motors	**~35-55+**	Window Lifts (4), Wipers (2), Coolant Pumps (3-4), Executive Seats (16-24 with memory/massage), Panoramic Roofs (2-4), Power Liftgate (2), Soft-close doors (4), Power steering column (2), etc. -
Medium Stepper Motors	**0**	No common applications identified. -
Total per Vehicle	**~55-85+**	-

Explanation of Differences Between Segments

The number of auxiliary motors in a BEV is directly proportional to its luxury level and feature content.

- **A/B-Segments:** These vehicles are designed for efficiency and affordability, so they have the fewest motors. They typically include essential motors for windows, wipers, and basic HVAC.
- **C/D-Segments:** As we move up the segments, we see more features like power seats, sunroofs, and advanced headlight systems, which all require additional motors. The increase in motor count is driven by the desire for more comfort and convenience.
- **E/F-Segments:** The luxury segments have the highest motor counts. This is due to the proliferation of advanced features like multi-zone climate control, massaging and ventilated seats, automatic doors, and active aerodynamic elements. In these vehicles, nearly every convenience feature is motorized, leading to a significant increase in the number of small and medium DC motors.

The transition from lower to higher segments shows a clear trend: as the price of the vehicle increases, so does the number of auxiliary motors. This is because manufacturers add more and more electrically-powered features to differentiate their premium models and provide a more luxurious experience for the occupants.